

SCD Type 2 dbt Snowflake CDC MERGE — parts A through D plus soft delete handling.

MERGE INTO DBT_DEV_DB.snapshots.snap_customers AS target
USING (

PART A DETECT CHANGED ROWS → CLOSE THE OLD VERSION

```
SELECT
    snap.dbt_scd_id,
    source.updated_at  AS dbt_valid_to,
    source.updated_at  AS dbt_updated_at,
    'update'          AS dbt_change_type
FROM (SELECT * FROM raw.public.customers) AS source
INNER JOIN DBT_DEV_DB.snapshots.snap_customers AS snap
    ON source.customer_id = snap.customer_id
    AND snap.dbt_valid_to IS NULL
WHERE source.updated_at > snap.dbt_updated_at

UNION ALL
```

PART B INSERT NEW VERSION OF CHANGED ROWS

```
SELECT
    MD5(
        COALESCE(CAST(source.customer_id AS VARCHAR), '')
        || '|'
        || COALESCE(CAST(source.updated_at AS VARCHAR), '')
    )          AS dbt_scd_id,
    NULL       AS dbt_valid_to,
    source.updated_at  AS dbt_updated_at,
    'insert'      AS dbt_change_type
FROM (SELECT * FROM raw.public.customers) AS source
INNER JOIN DBT_DEV_DB.snapshots.snap_customers AS snap
    ON source.customer_id = snap.customer_id
    AND snap.dbt_valid_to IS NULL
WHERE source.updated_at > snap.dbt_updated_at

UNION ALL
```

PART C INSERT BRAND-NEW ROWS (NOT YET IN SNAPSHOT)

```
SELECT
    MD5(COALESCE(CAST(source.customer_id AS VARCHAR), ''))
        AS dbt_scd_id,
    NULL       AS dbt_valid_to,
    source.updated_at  AS dbt_updated_at,
    'insert'      AS dbt_change_type
FROM (SELECT * FROM raw.public.customers) AS source
LEFT JOIN DBT_DEV_DB.snapshots.snap_customers AS snap
    ON source.customer_id = snap.customer_id
    AND snap.dbt_valid_to IS NULL
WHERE snap.customer_id IS NULL

UNION ALL
```

PART D SOFT DELETES → CLOSE ROWS WHERE IS_DELETED = TRUE

```
SELECT
    snap.dbt_scd_id,
    source.updated_at  AS dbt_valid_to,
    source.updated_at  AS dbt_updated_at,
    'delete'          AS dbt_change_type
FROM (SELECT * FROM raw.public.customers) AS source
INNER JOIN DBT_DEV_DB.snapshots.snap_customers AS snap
    ON source.customer_id = snap.customer_id
    AND snap.dbt_valid_to IS NULL
WHERE source.is_deleted = TRUE
```

) AS staged_changes
ON target.dbt_scd_id = staged_changes.dbt_scd_id

ACTION 1 MATCH + UPDATE OR DELETE → CLOSE THE OLD ROW

```
WHEN MATCHED
    AND staged_changes.dbt_change_type IN ('update', 'delete')
THEN UPDATE SET
    target.dbt_valid_to = staged_changes.dbt_valid_to
```

ACTION 2 NO MATCH + INSERT → ADD NEW / UPDATED ROW

```
WHEN NOT MATCHED
    AND staged_changes.dbt_change_type = 'insert'
THEN INSERT (
    customer_id, first_name, last_name, email,
    plan_type, city, is_active, is_deleted, updated_at,
    dbt_scd_id, dbt_updated_at, dbt_valid_from, dbt_valid_to
)
VALUES (
    staged_changes.customer_id,
    staged_changes.first_name,
    staged_changes.last_name,
    staged_changes.email,
    staged_changes.plan_type,
    staged_changes.city,
    staged_changes.is_active,
    staged_changes.is_deleted,
    staged_changes.updated_at,
    staged_changes.dbt_scd_id,
    staged_changes.dbt_updated_at,
    staged_changes.dbt_updated_at,
    NULL
);
```